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Kempegowda INTERNATIONAL AIRPORT BENGALURU

SECTION 03 30 00

CAST- IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 SUMMARY

A. General: Provide cast-in place concrete in accordance with requirements of the Contract Documents.

B. Related Work Specified Elsewhere

1. Section 01 81 13 “Sustainable Design Requirements.”
2. Section 01 44 00 “Construction Indoor Air Quality Management.”
3. Section 01 50 50 “Construction Waste Management.”
4. Section 01 81 15 “Volatile Organic Compound (VOC) Limits For Adhesives, Sealants, Paints & Coatings.”

C. Work Not Included

1. Plant and field inspection for concrete.

1.2 REFERENCES

A. General: Comply with the applicable provisions of the referenced standards except as modified by governing codes and the Contract Documents. All reference to Indian, British, American and other standards, regulations and requirements of statutory bodies and shall mean the latest published editions at the time of Contract. Where such standards, regulations and requirements are amended and affect the Contractor's responsibilities during the course of the Work, the Contractor shall immediately inform the Employer’s Representative (ER) of his intentions in respect of the revisions. Where a recommendation or suggestion occurs in the referenced standards, such recommendation or suggestion shall be considered mandatory. In the event of conflict between referenced standards, this specification or within themselves, the more stringent standard or requirement shall govern and as approved by the ER.

1. National Building Code of India.
2. Indian Standards

IS 269 Specification For Ordinary And Low Heat, Portland Cement.

IS 280 Specification For Mild Steel Wire For General Engineering Purpose

IS 303 Specification For Plywood For General Purpose.

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IS 383 Specification For Coarse And Fine Aggregates From Natural Sources For Concrete.

IS 432 Specification For Mild Steel And Medium Tensile Steel Bars And Hard Drawn Steel Wires For Concrete Reinforcement.
Part – 1 Mild steel and Medium tensile steel bars.
Part - 2 Hard drawn steel wire

IS 456 Code Of Practice For Construction And Design Of Reinforced Concrete

IS 460 Specification For Test Sieves:
(Part I, II & III)
Wire cloth test service
Perforated plate test sieve
Method of examination of test sieve

IS 516 Method Of Test For Strength Of Concrete

IS 650 - Standard sand for testing of cement

IS 723 Specification For Steel Counter Sunk Head Wire Nails.

IS 808 Dimensions For Hot Rolled Steel Beams, Channels And Angle Section

IS 814 Covered Electrodes For Metal Arc Welding Of Structural Steel.

IS 875 (Part 1) Code Of Practice For Design Loads (Other Than Earthquake)For Buildings And Structures Part 1 Dead loads - Unit weights of building material and stored materials (second revision)

IS 875 (Part 2) Code of practice for design loads (other than earthquake) for buildings and structures: Part 2 Imposed loads (second revision)

IS 875 (Part 3) Code of practice for design loads (other than earthquake) for buildings and structures: Part 3 Wind loads (second revision)

IS 875 (Part 5) Code of practice for design loads (other than earthquake) for buildings and structures Part 5 Special loads and load combinations (second revision)

IS 961 Specification For Structural Steel : High Tensile Steel Bars.

IS 1139 Hot Rolled M.S. Medium Tensile Steel And High Field Strength Deformed Bars For Concrete Reinforcement.

IS 1199 Method Of Sampling And Analysis Of Concrete.

IS 1343 Specification for Prestressed Design (24) SP 23 Handbook on concrete mix

IS 1387 General Requirements For Supply Of Metallurgical Materials.

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IS 1489 Specification For Portland Pozzolana Cement

IS 1542 Specification For Sand For Plaster

IS 1566 Hard Drawn Steel Wire Fabric

IS 1599 Method For Bend Test For Steel Products Other Than Sheets, Strip, Wire And Tube.

IS 1608 Method Of Tensile Testing: Steel Products

IS 1641:2013 Code Of Practice For Fire Safety Of Buildings (General): General Principles Of Fire Grading And Classification (Second Revision)

IS 1642:2013 Code Of Practice For Fire Safety Of Buildings (General): Details Of Construction (Second Revision) (Superseding IS 1645)

IS 1643:2013 Code Of Practice For Fire Safety Of Buildings (General): Exposure Hazard (Second Revision)

IS 1732 Dimensions For Round & Square Bars For Structural And General Engineering Purposes

IS 1786 Specification For Cold Worked Steel High Strength Deformed Steel Bars For Concrete Reinforcement (Superseding IS 1139)

IS 1791 - Batch type concrete mixers.

IS 1893(Part 1) Criteria for Earthquake Resistant Design of Structures: Part 1 General provisions and Buildings'

IS 1977 Specifications For Structural Steel : Ordinary Quality.

IS 2062 Specification For Structural Steel : Fusion Welding Quality.

IS 2116 Specification For Sand For Masonry Mortars

IS 2386 Method Of Test For Aggregate Concrete.
(Part I, II, & III)
Particle size and shape
Estimation of deleterious materials and organic impurities
Specific gravity, density, voids, absorption and bulking

IS 2502 Code Of Practice For Bending And Fixing Of Bars For Concrete Reinforcement

IS 2505 Immersion Type Concrete Vibrators

IS 2506 Screed Board Concrete Vibrators

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IS 2646 Specification For Integral Cement Water Proofing Compound.

IS 2722 Specification For Portable Swing Weigh Batcher (Single And Double Bucket Type)

IS 2751 Recommended Practice For Welding M.S Plain And Deformed .Bars

IS 3025 Methods Of Sampling And Test (Physical And Chemical For Water Used In Industry

IS 3068 Specification For Broken Brick (Burnt Clay) Coarse Aggregate For Use In Lime Concrete

IS 3366 Pan Vibrators

IS 3558 Code Of Practice For The Use Of Immersion Vibrators For Consolidating Concrete

IS 3696 Safety Code Of Scaffolds And Ladders :
Part 1 Scaffolds
Part 2 Ladders

IS 3809 Fire Resistance Test For Structures (First Revision)

IS 4014 (Parts 1 & 2) Code Of Practice For Steel Tubular Scaffolding.

IS 4031 Method Of Physical Test For Hydraulic Cement (Part I To XII)

IS 4032 Method Of Chemical Analysis For Hydraulic Cement.

IS 4082 Recommendation On Stacking And Storage Of Construction Materials At Site.

IS 4326 Earthquake Resistant Design and Construction of Buildings - Code of Practice

IS 4656 Form Vibrators For Concrete

IS 5640 Method Of Test For Determining Aggregate Impact Value Of Soft, Coarse Aggregate

IS 5816 Method Of Test For Splitting Tensile Strength Of Concrete Cylinder

IS 6452 Specification For High Alumina Cement For Structure Use.

IS 6461 Cement Concrete - Glossary Of Terms

IS 6909 Specification For Super Sulphated Weather

IS 7861 Code Of Practice For Extreme Weather Concreting
Recommended Practice For Hot Weather Concreting

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IS 8041 Specification For Rapid Hardening Portland Cement.

IS 8112 Specification For High Strength Ordinary Port Land Cement.

IS 8989 Safety Code For Erection Of Concrete Framed Structures

IS 9103 Specification For Admixture For Concrete.

IS 9417 Recommendations For Welding Cold Worked Steel Bars For Reinforced Concrete Construction.

IS 11433 Specification For One Part Gun Grade
Poly Sulphide Based Joint Sealant : General Requirements.

IS 12118 (Part I) Specification For Two Parts Poly Sulphate Based Sealant : General Requirements.

3. SP 24 Explanatory Handbook On Indian Standards Code For Plain And Reinforced Concrete (IS 456)

4. SP 27 Handbook Of Method Of Measurement Of Building Work.

5. British Standards

BS EN 197-1 Cement. Composition, specifications and conformity criteria for common cements

BS EN 12620 Aggregates for concrete (AMD Corrigendum 15333)

BS 1881 Methods of testing concrete.

BS EN 1008 Mixing water for concrete. Specification for sampling, testing and assessing the suitability of water, including water recovered from processes in the concrete industry, as mixing water for concrete.

BS 3892-1 Pulverized-fuel ash. Specification for pulverized-fuel ash for use with Portland cement.

BS 402 Specification for sulfate-resisting Portland cement

BS 444 Steel for the reinforcement of concrete. Weldable reinforcing steel. Bar, coil and decoiled product. Specification

BS 4483 Steel fabric for the reinforcement of concrete. Specification

BS 5075 Admixtures for concrete, mortar and grout.

BS 5328 Concrete. Specification, performance, production and conformity

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BS 8102 Code of practice for protection of structures against water from the ground

BS 8110-1 Structural use of concrete. Code of practice for design and construction

6. American Concrete Institute (ACI)

   a. ACI 207.1R – Mass Concrete.

   b. ACI 211.1 - Standard Practice for Selecting Proportions for Normal, Heavy Weight and Mass Concrete.

   c. ACI 301 "Specifications for Structural Concrete for Buildings".

   d. ACI 302.1R "Guide for Concrete Floor and Slab Construction".

   e. ACI 304.2R "Placing Concrete by Pumping Methods".

   f. ACI 305R "Hot Weather Concreting".

   g. ACI 308 "Standard Practice for Curing Concrete".

   h. ACI 309R "Guide for Consolidation of Concrete".

7. Concrete Reinforcing Steel Institute (CRSI)

   a. CRSI: "Manual of Standard Practice

   b. CRSI-WCRSI "Placing Reinforcing Bars".

8. United States Green Building Council (USGBC): Leadership in Energy and Environmental Design (LEED®) Green Building Rating System for Core and Shell Development (CS) Version 4.0 www.usgbc.org

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1.3 SYSTEM DESCRIPTION

A. Performance Requirements

1. Formwork Design: Design formwork, shoring and bracing to sustain construction and wind loads and maintain concrete lines and levels within concrete tolerances specified herein. Formwork types are labeled according to concrete surface finishes. Provide formwork types and construction to obtain the class of surface indicated and as required by referenced standards. Design forms to permit easy removal. Make the forms sufficiently rigid and strong enough to withstand, without leakage, the high hydraulic pressures which result from rapid filling and heavy high-frequency vibration. Deflection of each formwork component shall be limited to 1/360 of the component span due to hydraulic pressure of concrete.

B. Sustainable Design [and LEED] Performance Criteria

1. Credit MR 5, Regional Materials: The manufacturing locations and origin of raw materials for the cat-in-place concrete systems shall be identified and documented if within if sourced within a straight-line 800 kilometers total travel distance to the project site.

2. Credit MR 4, Recycled Content: The post-industrial and/or post-consumer recycled content (by weight) of the major cast-in-place concrete components shall be identified and documented.

    a. Recycled Content: Metal products shall strive to achieve a recycled content of 50% to 100%.

    b. Recycled Content: Concrete products shall strive to achieve a fly ash content of above 30%, but shall not exceed the requirements of IS 1489 (Part 1).

3. Credit MR 7, Certified Wood: Provide permanently installed wood-based materials and products that are a minimum 50% produced from wood obtained from forests and suppliers certified in accordance with FSC’s Principles and Criteria.

4. Credit IEQ 4.1, Low-Emitting Materials, Adhesives and Sealants: All field-applied adhesives, sealants, primers, paints and coatings used on the interior of the building shall meet the volatile organic compound (VOC) and chemical component limitations as defined in Section 01 81 15 Volatile Organic Compound (VOC) Limits For Adhesives, Sealants, Paints & Coatings" , VOC contents shall be identified and documented.

5. Credit IEQ 4.2, Low-Emitting Materials, Paints and Coatings: All field-applied paints and coatings used on the interior of the building shall meet the volatile organic compound (VOC) and chemical component limitations as defined in Section 01 81 13 "Sustainable Design Requirements". VOC contents shall be identified and documented.

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6. Credit IEQ 4.4, Composite Wood and Agrifiber Products: All composite wood and agrifiber products used on the interior of the building shall meet the volatile organic compound (VOC) and chemical component limitations as defined in Section 01 81 13 "Sustainable Design Requirements" and contain no added urea-formaldehyde resins. VOC contents shall be identified and documented.

7. Credit MR 7, Certified Wood: Provide permanently installed wood-based materials and products that are a minimum 50% produced from wood obtained from forests and suppliers certified in accordance with FSC's Principles and Criteria.

B. Design Modifications: Make design modifications of Work shown only as may be necessary to meet performance requirements and coordinate the Work. Variations in details and materials which do not adversely affect appearance, durability or strength shall be submitted to the ER for review. Maintain the general design concept without altering profiles and alignments shown.

1.4 SUBMITTALS

A. Product Data: Submit for Employer's Representative's (ER's) action. Submit product data for proprietary materials and items, including reinforcement and forming accessories, cement, aggregate, admixtures, patching compounds, waterstops, joint systems, curing compounds, finishing materials and others if requested by ER.

1. Form Release Agent: Furnish data stating that the product will not stain the concrete surfaces and will not adversely affect the bond of subsequently applied finishes.

B. Shop Drawings: Submit for ER's action. Submit shop drawings for the fabrication and installation of the Work. Prepare details at not less than 1:15 minimum scale.

1. Steel Reinforcement Shop Drawings: Provide, prior to fabrication, details of fabrication, bending, and placement, prepared according to ACI 315, "Details and Detailing of Concrete Reinforcement." Include material, grade, sizes, bar schedules, stirrup spacing, bent bar diagrams, arrangement, and supports of concrete reinforcement. Include special reinforcement required for openings through concrete structures.

2. Detailed drawings showing locations of construction joints, slab edges, curbs, equipment pads, depressions, sleeves, openings and the locations of adjacent Work required by other trades.

3. Setting drawings showing layout of inserts and block-outs for the Work of other Sections.

4. Shop drawings shall not be, nor shall they incorporate, reproductions of Contract Documents or portions thereof.

C. Formwork Placement Drawings: Submit for ER's information and for Governmental Agencies as requested. Provide schematic formwork placement drawings showing pre-shoring, shoring and reshoring with sequential outline describing stripping times and reshoring placement. Include descriptive literature for formwork materials. Design and engineering of formwork and formwork placement drawings are the Contractor's responsibility.

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D. Calculations

1. Reshoring Calculations: Submit, for ER's information, written and diagrammatic description of shoring/reshoring timeline procedure including the number of floors to receive re-shores with temporary construction loads.

E. Construction Joint Layout: Submit for ER's information. Provide layout drawings indicating proposed construction joints required to construct the structure. Location of construction joints is subject to approval of the ER.

F. Quality Control Submittals: Submit for ER's information.

1. Materials:

a. Concrete Mix Designs: Certified report identifying the design mixes, stating where each mix shall be used, the mix proportions, and additional design information including adjustments for seasonality or flowability of concrete. Submit mix design forms including not less than the following items:

1) Project Data: Identify name, location, Contractor, concrete strength (class), and concrete use (describe).

2) Basis of Design: Compliance with mix design requirements shall be substantiated by:

a) Specified grade < M70 (fck < 70 MPa): Field test records, trial batches, or calculations prepared by and containing the seal of a qualified Professional Engineer licensed in India. (Submit Professional Engineer's credentials and license).

3) For test records or trial batches, statistical procedures for required average strength shall be utilized. Submit calculations for review with mix design.

4) Design Characteristics: List mix density, characteristic strength—28-day strength (or 56-day as required) and air content.

5) Mix Proportions: List each material with type/source, specific gravity, weight, absolute volume, mix density (total of weight/volume) and mix water-cementitious ratio.

2. Procedures:

a. Flatwork Finishing: Furnish a complete description of the system proposed for meeting the specified floor slab flatwork tolerances. Include evidence of previous successful use of the proposed system.

b. Curing Procedures: Furnish a written description of the curing procedures and methods proposed for use in the Work. Include a description of the tests to be used for determining the duration of the curing period.

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3. Tests Reports:

   a. Furnish reports for the specified Source Quality Control and Field Quality Control inspections and tests. Test reports indicating non-compliance should be immediately forwarded with an appropriate corrective action proposed.

   b. Mill Tests: Certified mill test reports for cement, metal reinforcement and welded wire fabric.

4. Certificates

   a. Certifications for Admixtures: Each admixture manufacturer shall submit certification that admixture utilized in the Work complies with specified requirements and to the appropriateness of the use of their admixtures in the combination, dosages, batching, and construction procedures proposed. In addition each admixture manufacturer shall certify that their admixture is non-corrosive based upon long term testing and compatible with other components proposed mixes.

5. Qualifications

   a. Installer's Qualifications: Submit documentation identifying Contractor's Qualifications comply with specified requirements. Submit evidence of successful completion of work of similar scope to that shown and specified for this Project.

G. Green Building Submittal Requirements

1. A completed GREEN BUILDING MATERIALS CERTIFCATION FORM (GBMCF), appended to Section 01 81 13: "Sustainable Design and Construction". Information to be supplied for this form shall include:

   a. Cost breakdowns for the materials included in the Contractor or sub-Contractor's Work. Cost breakdowns shall include total cost plus itemized material costs for the cast-in-place concrete systems and associated products.

   b. The manufacturing location and raw material sources of the cast-in-place concrete and associated products.

   c. The amount of post-consumer and/or post-industrial recycled content in the cast-in-place concrete systems.

   d. The VOC content of all interior and field-applied adhesives, sealants, paints, and coatings.

2. GBMCF Back-Up Documentation: These documents are used to validate the information provided on the GBMCF (except cost data). For each material listed on the GBMCF, provide documentation to certify the material's GREEN BUILDING attributes, as applicable:

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a. Credit MR 4, Recycled Content: Recycled content: Provide published product literature or letter of certification on the manufacturer's letterhead certifying the amounts of post-consumer and/or post-industrial content.

b. Credit MR 5, Regional Materials: Regional manufacturing (within (800 kilometers): Provide published product literature or letter of certification on the manufacturer's letterhead indicating the city/state from which each of the raw materials in the product were extracted, harvested or recovered, and the distance in miles from the project site. If only some of the raw materials for a particular product or assembly originate within (800 kilometers) of the project site, provide the percentage (by weight) that these materials comprise in the complete product. t site.

c. Credit MR 7, Certified Wood: The percentage of new wood that is to be FSC certified, and availability of Chain-of-Custody certificates. Include statement indicating projected costs for all new wood products.

d. Credit IEQ 4.1, Low-Emitting Materials, Adhesives and Sealants: The VOC content of all interior and field-applied adhesives and sealants.

e. Credit IEQ 4.2, Low-Emitting Materials, Paints and Coatings: Submit printed documentation demonstrating compliance with specified performance requirements.

f. Credit IEQ 4.4, Low-Emitting Materials, Composite Wood and Agrifiber Products: The VOC content of all interior and field-applied composite wood and agrifiber products.

g. Product cut sheets for materials that meet the GREEN BUILDING Materials Certification Form. Cut sheets shall be submitted with the Contractor or sub-Contractor's stamp, as confirmation that the submitted products are the products installed in the Project

h. Material Safety Data Sheets (MSDS), for applicable products. Applicable products include, but are not limited to field-applied adhesives, sealants, carpets, paints and coatings used on the interior of the building. Material Safety Data Sheets shall indicate the Volatile Organic Compound (VOC) limits of products submitted (If an MSDS does not include a product's VOC limits, then product data sheets, manufacturer literature, or a letter of certification from the manufacturer can be submitted in addition to the MSDS to indicate the VOC limits).

3. The GREEN BUILDING submittal information shall be assembled into one (1) package per Section or trade, and sent to the Consultant of review. Incomplete or inaccurate GREEN BUILDING submittals may be used as the basis for rejecting the submitted products or assemblies.

1.5 QUALITY ASSURANCE

A. Basis of Design: When particular manufacturers' materials, products or processes are specified for an item of Work, any one thereof is acceptable for the Contractor to choose. An

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alternative material, product or process will be considered if the Contractor submits a written substitution request together with such information as may be necessary to assist the ER in determining whether the proposed substitution is acceptable; the burden of proof rests solely upon the Contractor.

B. Qualified Cast-in-Place Concrete Installer: The cast-in-place concrete work shall be performed by a firm having 5 years' experience in the installation of specified materials on comparable projects. The installer shall provide evidence of successful completion of Work of similar scope to that shown and specified for this Project using similar cast-in-place concrete systems and whose Work has resulted in construction with a record of successful in-service performance.

C. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for formwork and shoring and reshoring installations that are similar to those indicated for this Project in material, design, and extent.

D. Sole Source Responsibility: Obtain each type of class of cementitious material of the same brand from the same manufacturer's plant, each aggregate from one source, and each admixture from the same manufacturer for each type specified.

E. Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products requirements for production facilities and equipment.

F. Welding Qualifications: Qualify procedures and personnel according to IS 2751 "Recommended Practice For Welding M.S Plain And Deformed .Bars " and AWS D 1.4M, "Structural Welding Code - Reinforcing Steel".

G. Regulatory Requirements: Comply with applicable requirements of the laws, codes, ordinances and regulations of authorities having jurisdiction. Obtain necessary approvals from authorities

H. Pre-Construction Conference: At least 35 days prior to submitting the proposed mix designs, conduct a conference to review the proposed materials, mix designs, construction procedures and finishing systems for the concrete work. Topics to include but are not limited to pour sequence, trade placement sequence, shoring/reshoring sequence, slab surveys, and highlight of all requirements the Contractor or subcontractor has taken exception to. The meeting shall be attended by the ER and representatives of the Testing Agencies and the subcontractors involved with supplying, conveying, placing, finishing, and curing the concrete. Furnish the conference agenda to attendees 20 days prior to the scheduled conference date. Meeting shall include the following:

1. Employer's Representative.
2. Engineer in Charge (ER).
3. General Contractor.
4. Concrete Subcontractor.

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5. Independent Testing Agancy.

6. Ready mix concrete supplier.

1.6 DELIVERY, STORAGE AND HANDLING

A. Packing, Shipping, Handling, and Unloading: Use materials from the same source from start to finish of job unless changes have been accepted by the ER. Deliver and store materials in manufacturer's original packaging labeled to show name, brand, type, and grade. Store, materials in protected dry location off ground in accordance with manufacturer's instructions. stockpile and use aggregate in a manner which minimizes moisture content variation. Concrete producer and suppliers shall load, transport, handle, store and batch concrete materials in a manner that will ensure the materials are not contaminated, unclean, segregated or affected in a way that will damage the final product.

1. Bagged Cement: Store on raised platforms in a dry area. Use bagged cement on the basis of first received, first used. Do not store bagged cement over 6 months.

2. Reinforcement: Deliver reinforcement to the job site bundled, tagged and marked. Use metal tags indicating bar size, lengths and other information corresponding to markings shown on shop drawings. Store reinforcement at the job-site in a manner to prevent damage and accumulation of dirt and excessive rust.

1.7 PROJECT/SITE CONDITIONS

A. Hot or Cold Weather Concreting: Comply as follows:

1. During hot or cold weather the concreting should be done as per the procedures set out in IS 7861 Part I or IS 7861 Part II or as directed by the ER.

2. Review with the ER the methods of protection and/or mix design modifications to be used for hot weather concreting. This review shall occur prior to the mean daily temperature exceeding (27 deg. C.)

3. When the temperature is above (32 deg. C.), reduce the mixing/delivery time from 1-1/2 hrs. to 1-1/4 hrs. Cool forms to a maximum of (27 deg. C.) Cool concrete to maximum of (21 deg. C.) leaving the mixer.

4. Do not place concrete at a temperature greater than (35 deg. C.)

5. When ambient placement temperatures are more than (27 deg. C.), provide water-reducing, set-retarding admixture in accordance manufacturer's recommendations

B. Concreting In Aggressive Soils and Water:

1. Guidelines laid down in 13.4 of IS 456 shall be followed together with the instruction of the ER.

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PART 2 - PRODUCTS

2.1 FORMWORK MATERIALS

A. Formwork: Provide form-facing panels that will provide continuous, true, and smooth (unless otherwise specified) concrete surfaces. Furnish in largest practicable sizes to minimize number of joints. Utilize materials for formwork that will produce desired results and as approved by the ER.

B. Form Ties: For securing forms where surfaces will be exposed in the finished work, use tie screws with removable plastic cones, removable bolts, special removable tie wires or Series 300 stainless steel snap ties designed to resist lateral pressure of fresh concrete on forms and to prevent spalling of concrete on removal. For other forms, use either bolts or wires. Use ties of type that when forms are removed, no metal is closer than (38mm) from the finished concrete surface. Furnish ties that, when removed, will leave holes no larger than (25mm) in diameter in concrete surface. Furnish ties with integral water-barrier plates to walls indicated to receive damp-proofing or waterproofing.

C. Form Coatings (Release Agent): Provide commercial formulation form-coating compounds with a maximum that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces. Formulate form-release agent with rust inhibitor for steel form-facing materials.

D. Blockouts, Keyways: Wood or extruded closed–cell polystyrene insulation.

E. Chamfer Strips: Wood, polyvinylchloride, or neoprene. Compatible with finish.

F. Wood Rustication Strips: Milled straight, kerfed on the back face, without protruding splinters. Fully seal before application to the forms.

G. Wood Nailing Strips: Solid wood stock, preservative treated i Minimum retention of (4kg/m^3) preservative of wood.

H. Anchoring Inserts: Provide proprietary type inserts and applicable installed spacings as indicated on the Drawings.

2.2 REINFORCING STEEL BARS, WELDED STEEL WIRE FABRIC AND ACCESSORIES

A. Reinforcing Steel: Plain Bars (Only Where Shown or Noted).

1. Rolled mild steel and medium tensile steel plain round bars used in concrete shall conform to IS 432 Part I. Steel received shall conform to the following IS with regards to manufacturing and chemical composition.

    a. M.S. bar Grade I — Steel designation Fe 500-S of IS 226

    b. M.S. bar Grade II — Steel designation Fe 500-O of IS 1977

    c. Medium Tensile Steel designation Fe 540 W-HT IS 961 Steel bars

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2. National sizes and tolerances shall be as specified in IS 432 Part I. Physical requirements shall be determined in accordance with IS 1608, read in conjunction with IS 226 and shall be as reproduced herein as Table 1 for ready reference.

B. Reinforcement Steel: Deformed Bars (Typical, Unless Noted Otherwise)

1. High strength deformed bars for use as reinforced in concrete shall be of grade Fe 415, Fe 500 and Fe 550 conforming to IS 1786.

2. Chemical composition shall conform to IS 1786 when made as a relevant part of IS 228. Permissible limits shall be as reproduced herein as Table 2 for ready reference.

3. Welding of cold worked steel bars in reinforcement shall be permitted as per IS 9417 (Recommendation for welding cold worked steel bars for RCC)

4. Nominal sizes, cross sectional areas and their mass shall be as specified in IS 1786, allowing due consideration for tolerance specified therein.

C. Physical properties

1. Bars shall satisfy IS 1599 test for bend and re-bend test in conjunction with IS 226.

2. Bond requirements shall be deemed to have been satisfied if bars meet requirements of clause 4.0 of IS 1786.

3. Tensile, proof stress and percent elongation shall be as per Table 3 of IS 1786 and reproduced herein as Table 3 for ready reference.

D. Material received at site shall have ISI certification mark. Each bundle or coil containing the bars shall be suitably marked with ISI certification mark. Also bars shall be marked to identify categories. This shall be done as per IS 1387.

E. In case bars are without ISI certification mark, the manufacturer shall provide a certificate stating process of manufacturer, chemical composition and mechanical properties. Each certificate shall indicate the number or identification mark of the batch production / cast to which it applied. Corresponding number or identification mark should be found on the material.

F. All reinforcement material shall be free from loose mill scale, excessive rust, loose rust, pitting, oil, grease, paint, mud or any foreign deleterious material present on the surface. Cleaning shall be done to the satisfaction of the ER.

G. Each batch brought at site shall be tested prior to use for respective specification / Physical properties. Cost of all such tests shall be borne by the Contractor. Material acceptable as per IS shall be allowed into the Work. All rejected material shall be removed from site by the Contractor within 3 days of rejection.

H. Reinforcement bars received at site shall be stored on hard concrete platform and clear of the ground with the use of timber sleeper, concrete sleeper or any other means. Reinforcement material shall be kept covered by tarpaulins or plastic to avoid corrosion and other contamination. It is advised to follow storage methods as described in IS 4082.

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TABLE – 1

MECHANICAL PROPERTIES OF MILD STEEL & MEDIUM TENSILE STEEL BARS

[IS 432 Table – 1]

SR. NO.	TYPE AND NOMINAL SIZE OF BARS	ULTIMATE TENSILE STRESS	YIELD STRESS	ELONGATION* PERCENT
		MIN.	MIN.	MIN.
1.	Mild Steel Grade I			
	For bars up to and including 20 mm	410	250	23
	For bars over 20 mm, upto and including 50 mm	410	240	23
2.	Mild Steel Grade II			
	For bars up to and including 20 mm	370	225	23
	For bars over 20 mm, upto and including 50 mm	370	215	23
3.	Medium Tensile Steel			
	For bars up to and including 16 mm	540	350	20
	For bars over 16 mm, upto and including 32 mm	540	340	20
	For bars over 32 mm, upto and including 50 mm	510	330	20

* Elongation on a gauge length 5.65 √So where So is the cross sectional area of the test piece.

TABLE – 2

CHEMICAL COMPOSITION OF HIGH STRENGTH DEFORMED BARS

[IS 1786]

Constituent	For ladle analysis of steel when made as per relevant parts of IS 228 Percent Maximum*			For product analysis Variation, Over Specified Maximum Limit, Percent, Max
	Fe 415	Fe 500	Fe 550	Limit % Maximum
Carbon	0.30	0.30	0.30	0.02
Sulphur	0.060	0.055	0.055	0.005
Phosphorus	0.060	0.055	0.050	0.005
Sulphur & Phosphorus	0.11	0.105	0.10	0.010

* NOTE – 1 For guaranteed weld ability, the percentage of carbon shall be restricted to 0.25 percent maximum.

* NOTE – 2 Addition of micro alloying elements is not mandatory for any of the above grades. When strengthening elements like Nb, V, B and Ti are used individually or in combination, the total contents shall not exceed 0.30 percent; in such case manufacturer shall supply the purchaser or his authorized representative a certificate stating that the total contents of the strengthening elements in the steel do not exceed the specified limit.

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TABLE – 3

MECHANICAL PROPERTIES OF HIGH STRENGTH DEFORMED BARS AND WIRES

[IS 1786 Table – 3]

SR. NO.	PROPERTY	GRADE		
		Fe 415	Fe 500	Fe 550
(1)	(2)	(3)	(4)	(5)
i)	0.2 percent proof stress/yield stress, Min, N/Sqmm.	415.0	500.0	550.0
ii)	Elongation, percent, Min, on gauge length 5.65 VA, where A is the cross sectional area of the test piece.	14.5	12.0	8.0
iii)	Tensile strength, Min	10 percent more than the actual 0.2 percent proof stress but not less than 485.0 N/Sqmm.	8 percent more than the actual 0.2 percent proof stress but not less than 545.0 N/Sqmm.	6 percent more than the actual 0.2 percent proof stress but not less than 585.0 N/Sqmm.

2.3 GRADES OF CONCRETE

A. Concrete Grade Designations and corresponding required characteristic compressive strength in the Work shall be as shown in Table 4.

TABLE -- 4

CONCRETE GRADES AND CHARACTERISTIC COMPRESSIVE STRENGTHS

[IS 456 – TABLE 2]

GRADE DESIGNATION	SPECIFIED CHARACTERISTIC COMPRESSIVE STRENGTH [N/mm^2] AT 28 DAYS
1	2 [N/mm^2]
M 10	10
M 15	15
M 20	20
M 25	25
M 30	30
M 35	35
M 40	40
M 60	60

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NOTE – 1 In the designation of a concrete of mix, letter M refers to the mix and the number to the specified characteristic compressive strength of 15 cm cube at 28 days, expressed in N/mm^2.

NOTE – 2 M 5 and M 7.5 grade of concrete may be used for lean concrete bases and simple foundations for masonry walls. These mixes need not be designed.

NOTE – 3 Grades of concrete lower than M 15 shall not be used in reinforced concrete.

2.4 DURABILITY AND COMPATIBILITY REQUIREMENTS FOR CONCRETE AND MATERIALS

A. General: Use materials and produce concrete which comply with the following requirements in Tables 5 and 6:

TABLE – 5

MINIMUM CEMENT CONTENT REQUIRED IN CEMENT CONCRETE TO ENSURE DURABILITY UNDER SPECIFIED CONDITIONS OF EXPOSURE

[SP – 23 – Table 23 & IS 456 – Table 19]

EXPOSURE	PLAIN CONCRETE		REINFORCED CONCRETE	
	MINIMUM CEMENT CONTENT	MAXIMUM WATER CEMENT RATIO	MINIMUM CEMENT CONTENT	MAXIMUM WATER CEMENT RATIO
(1)	(2)	(3)	(4)	(5)
	Kg/m^3		Kg/m^3	
Mild – for example, completely protected against weather, or aggressive conditions, except for a brief period of exposure to normal weather conditions during construction.	220	0.7	250	0.65
Moderate – for example, sheltered from heavy and wind driven rain and against freezing, whilst saturated with water; buried concrete in soil and concrete continuously under water.	250	0.6	290	0.55
Severe – For example, exposed to sea water, alternate wetting and drying and to freezing whilst wet, subject to heavy condensation or corrosive fumes.	310	0.5	360	0.45

NOTE – 1 When the maximum water cement ratio can be strictly controlled, the cement content in the above table may be reduced by 10 percent.

NOTE – 2 The minimum cement content is based on 20 mm, aggregate. For 40 mm aggregate, it should be reduced by about 10 percent; for 12.5 mm aggregate, it should be increased by about 10 percent.

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TABLE – 6

REQUIREMENTS FOR CONCRETE EXPOSED TO SULPHATE ATTACK

[SP – 23 – Table 24 & IS 456 – Table 20]

CLASS	CONCENTRATION OF SULPHATES EXPRESSED AS SO_3			TYPE OF CEMENT	REQUIREMENTS FOR DENSE FULLY COMPACTED CONCRETE MADE WITH AGGREGATE COMPLYING WITH IS : 383*	
	IN SOIL		IN GROUND WATER			
	TOTAL SO_3 (PERCENT)	SO_3 IN 2:1 WATER EXTRACT g/l	(PARTS PER 100 000)		MINIMUM CEMENT CONTENT	MAXIMUM FREE WATER / CEMENT RATIO
(1)	(2)	(3)	(4)	(5)	(6)	(7)
1)	Less than 0.2	--	Less than 30	Ordinary Portland cement or Portland slag cement or Portland Pozzolana cement	280 kg/m^3	0.55
2)	0.2 to 0.5	--	30 to 120	Ordinary Portland cement or Portland slag cement or Portland Pozzolana cement Super sulphated cement	330 kg/m^3 310 kg/m^3	0.50 0.50
3)	0.5 to 1.0	1.9 to 3.1	120 to 250	Super sulphated cement	330 kg/m^3	0.50

2.5 CEMENT MATERIALS

A. Cement shall be ordinary Portland cement conforming to IS 269 for all purpose unless specified differently. It shall be received in bags of 50 kg and each batch shall be accompanied with a test certificate of the factory. Also it shall be tested before use to ascertain its strength, setting time, etc. In no case shall cement that has been stored over 4 weeks be used unless tested as per the direction of the ER prior to use in the Work.

B. Cement shall be stored in such locations so as to prevent deterioration due to moisture dampness. A dry and water proof shed shall be best suited for this. Bags shall be stacked on rigid water-proof platforms about 15 to 20 cm clear above the floors and 25 to 35 cm clear or away from the surrounding walls. The stack height shall not exceed of 12 bags. Stacks shall be so arranged that the first batches are used first, and (FIFO) that they permit easy access for inspection and handling.

C. The following other types of cement may be used in Work if specified or with prior approval of the ER in writing purpose. Specialist literature shall be consulted for guidance regarding use of these types of cement.

1. 43 Grade ordinary Portland cement conforming to IS 8112
2. 53 Grade ordinary Portland cement conforming to IS 12269
3. Portland slag cement conforming to IS 455

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4. Portland pozzolana cement (fly ash based) conforming to IS 1489 (Part-I)

5. Portland pozzolana cement (calcined clay based) conforming to IS 1489 (Pt –2)

6. Sulphate resisting Portland cement conforming to IS 12330

2.6 WATER

A. Water used for mixing and curing shall be clean, reasonably clear and free from harmful quantities of silt, oils, alkalis, acids, salts, sugars, and organic matter that could be detrimental to the concrete or reinforcing steel, cause efflorescence / stains / deposits, affect the hydration process or otherwise interfere with hardening of concrete during curing. Potable water is generally considered satisfactory for use but shall be tested prior to use in the Work.

B. Water shall be tested in accordance with IS 3025. Maximum permissible limits of deleterious materials in water shall be as given in IS 456 and Table --7.

C. Suitability of water shall be ascertained by the compressive strength and initial setting time test as specified under:

1. Average 28 days compressive strength of at least three 15 cm concrete cubes prepared with water proposed to be used shall not be less than 90% of the average strength of three similar concrete cubes prepared with distilled water. Preparation and testing in accordance of IS 516.

2. The initial setting time of tests blocks made with proposed cement and water to be used shall not be less than 30 minutes and shall not differ by ± 30 minutes from the initial setting time of control test block prepared with the same cement and distilled water. Preparation and testing of block shall be in accordance with IS 4031

3. The PH value of water shall not be less than 6 and more than 9.

D. Water storage tanks shall be such as to prevent any deleterious materials getting mixed into concrete.

E. Water tested as above shall be approved in writing by the ER prior to use in the Work.

TABLE – 7

PERMISSIBLE LIMITS FOR SOLIDS (IN WATER)

	Permissible Limit, Maximum
Organic	200 mg/l
Inorganic	3000 mg/l
Sulphates (as SO_4)	500 mg/l
Chlorides (as C1)	2000 mg/l for plain concrete work and 1000 mg/l for reinforced concrete work
Suspended matter	2000 mg/l

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2.7 ADMIXTURES FOR CONCRETE

A. General: Admixtures shall conform to the requirements of IS 9103, ASTM C260, ASTM C10, ASTM C 595 or ASTM C 618. Use shall conform to the requirements of IS 456 and IS 9103. Calcium chloride and thiocyanates are not permitted. Provide only those admixtures which produce no detrimental effects on metal components such as form and composite decking, reinforcing steel, embedments and conduit. Admixtures, when combined with other ingredients used to produce the concrete, shall result in concrete having not more than the limits for soluble chloride ion content in the specified standards or 0.6 kg/m^3 of concrete. All structural concrete shall contain a water reducing admixture.

B. Admixtures generally in use are classified as follows:

1. Accelerators
2. Retarders
3. Workability agents
4. Water -repelling agents
5. Air-entraining agents
6. Gas-forming agents.

C. Admixtures are manufactured and sold under trade names. Product data for each admixture proposed to be used in the Work shall indicate its trade name, chemical composition, purpose of use, source, directions for use, dosage rates and the possible detrimental effects of under and over-dosage. It shall also indicate chloride and sulphate content. Product data for air entraining admixtures shall indicate the average expected air content in freshly mixed concrete containing when dosed at the manufacturer's recommended rate..

D. Fly Ash: Shall conform to IS 3812 (Parts I and II) or ASTM C618, Class C or F except limit lignite ignition to maximum of 6% with maximum of (220kg/m^3). No additives or contaminants derived from the fly ash precipitation process that are detrimental to concrete (such as soda ash) are permitted. The fly ash producer shall have a minimum of 5 years' experience in the production of acceptable fly ash and shall maintain an effective quality control program to guard against contamination of the fly ash.

E. Other Admixtures: Use only with the prior written approval of the ER. Do not use any admixtures which contain chlorides or sulphates.

F. Compatibility of multiple admixtures, including fly ash, used in the same mix shall be certified

2.8 AGGREGATES FOR CONCRETE

A. Coarse Aggregate

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1. Coarse aggregate shall be natural materials such as stone, gravel etc—crushed, uncrushed or a combination thereof—obtained from approved quarries. Aggregate shall be hard, strong, dense, durable, clean and free from veins and adherent coating. It shall be free from soft, friable, flat, thin, elongated or laminated pieces and shall be roughly cubical in shape. It shall consist of coarse material most of which is retained on 4.75 mm IS sieve.

2. Coarse aggregate shall not contain any harmful material such as iron, pyrites, coal, mica shale or similar laminated material or clay, alkali, soft fragments, sea shells, organic impurities etc. in quantities that can adversely affect the strength and durability of the concrete. In addition to the above, in reinforced concrete the aggregate shall not contain any material that might attack the reinforcement. The maximum quantities of deleterious materials in the coarse aggregate when determined in accordance with IS 2386 Part I and Part II “Method of test for aggregates for concrete’ shall not exceed the limits provided in Table -- 8.

3. Aggregate crushing value, impact value, abrasion value and soundness of aggregate shall respectively be in accordance with paragraphs 3.3, 3.4, 3.5 and 3.6 of IS 383.

4. Grading of coarse aggregate shall conform to the requirements in IS 383. See Table – 9.

5. Source of aggregate shall be an approved Government location. It shall be tested prior to the approval of the ER by an approved testing laboratory. In case available aggregates do not meet specific requirements of IS 383 or any other applicable specification, required processing shall be carried out by the Contractor at his cost. It shall be the duty of the Contractor to make sure that aggregate material received by him is from Government approve quarries and with fully paid royalties, taxes, duties etc. as may be in force from time to time for respective locations.

6. Aggregate shall be stored in such a way that it does not get mixed with mud, grass vegetables and other foreign matter. The best way ensure this is to have a hard surface platform made out of concrete, bricks or planks. It should meet the approval of the ER.

7. Coarse aggregate shall have a minimum specific gravity of 2.6 (Saturated surface dry basis). Aggregate below this specific gravity shall not be used without the special permission of the ER.

8. Once a specific source of supply of coarse aggregate is approved, the source shall not be changed without the prior approval of the ER.

B. Fine Aggregate

1. Natural sand deposited by stream or glacial agencies as a result of disintegration of rock is the best form of fine aggregate. The fine aggregate shall confirm to following standards.

   a. For Plain And Reinforced Concrete: IS 383 Specification for coarse and fine aggregates from natural sources for concrete.

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2. Can be obtained from crushed stone screening but often contains a high percentage of dust and clay. It tends to be flaky and angular. This type produces harsh concrete and should be avoided.

3. Sea sand shall not be used unless approved by the ER. If approved, the required treatment shall be done at the Contractor's cost.

4. Sand shall be hard, durable, clean and free from adherent coatings and organic matter and shall not contain any appreciable amount of clay. Sand shall not contain harmful impurities such as iron, pyrites, coal particles, lignite, mica shale or similar laminated material, alkalis, and organic impurities in such form or quantity as to affect the strength or durability of concrete. It should also not contain any material liable to attack the steel reinforcement.

5. When tested per IS 2386 Part I and Part II, deleterious materials in fine aggregate shall not exceed permissible limits given in Table -- 8.

6. Fine aggregate shall be thoroughly washed at site with clean fresh water such that the percentage of all deleterious materials is within the permissible limits laid down.

7. Grading of fine aggregate shall conform to IS 383 and shall fall within limits of one of the four zones given in table 4 of IS 383. See Table -- 10.

8. Damp and moist sand increases in volume and this effect is called bulking. Due allowance shall be made for this effect when preparing mixes based on volume measurements. Bulking effects shall be determined per IS 2386 Part III Appendix A. For rough guidance Table --12 gives the relation between moisture content and percentage of bulking.

9. Storing of aggregate shall be as given in clause 2.8.6.

C. Aggregate Usage: Nominal aggregate size for different types of concrete usage shall be as follows.

1. General Use: Size 20.

2. Footings, Mat Foundations, Grade Beams, and Walls Over 300 mm thick: Size 40

3. Tight Pours: Size 12.5 or Size 10.

4. Non-Structural Concrete Fill, Column Encasement, and Fill: Size 12.5.

5. Fine Aggregate for Concrete Fill: 100% passing the 4.75 mm sieve and maximum 5% passing the 0.15mm sieve.

6. Sand Fineness Modulus (FM): Minimum 2.50, maximum 3.00.

D. All-In Aggregate Grading: See Table 11 for all-in aggregate grading for nominal sizes 40 and 20.

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TABLE – 8

LIMITS OF DELETERIOUS MATERIALS (AGGREGATES)

[IS 383 – Table 1]

NO.	DELETERIOUS SUBSTANCE	METHOD OF TEST	FINE AGGREGATE PERCENTAGE BY		COARSE AGGREGATE PERCENTAGE BY	
(1)	(2)	(3)	(4) Weight, Max Uncrushed	(5) Weight, Max Crushed	(6) Uncrushed	(7) Crushed
i)	Coal and lignite	IS : 2386 (Part – II)	1.00	1.00	1.00	1.00
ii)	Clay Lumps	- do -	1.00	1.00	1.00	1.00
iii)	Material finer than 75 – (MU) IS Sieve	IS : 2386 (Part – I)	3.00	15.00	3.00	3.00
iv)	Soft fragments	IS : 2386 (Part – II)	--	--	3.00	--
v)	Shale	IS : 2386 (Part – II)	1.00	--	--	--
vi)	Total of percentage of all deleterious materials (except mica) including Sl. No. (i) to (v) for Col. 4,6 and 7 and Sl. No. (i) and (ii) for Col. 5 only	--	5.00	2.00	5.00	5.00

Note :-1The Presence of mica in the fine aggregate has been found to reduce considerably the durability and compressive strength of concrete and further investigations are underway to determine and extent of the deleterious effect of mica. It is advisable, therefore, to investigate the mica content of fine aggregate and make suitable allowance for the possible reduction in the strength of concrete mortar.

Note :-2The aggregate shall not contain harmful organic impurities (tested in accordance with IS : 2386 [part – ii]) in sufficient quantities to affect adversely the strength or durability of concrete. A fine aggregate which fails in the test for organic impurities may be used, provided that, when tested for the effect of organic impurities on the strength of mortar, the relative strength at 7 and 28 days, reported in accordance with 7 of IS : 2386 [Part – VI] – 1963 is not less than 95% percent.

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TABLE – 9

GRADING OF COARSE AGGREGATES

[IS 383 – Table 2]

IS Sieve Designation	Percentage passing for single – sized Aggregate of Nominal Size						Percentage passing for Graded Aggregate of Nominal Size			
	63 mm	40 mm	20 mm	16 mm	12.5 mm	10 mm	40 mm	20 mm	16 mm	12.5 mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
80 mm	100	--	--	--	--	--	100	--	--	--
63 mm	85 to 100	100	--	--	--	--	--	--	--	--
40 mm	0 to 30	85 to 100	100	--	--	--	95 to 100	100	--	--
20 mm	0 to 5	0 to 20	85 to 100	100	--	--	30 to 70	95 to 100	100	100
16 mm	--	--	--	85 to 100	100	--	--	--	90 to 100	--
12.5 mm	--	--	--	--	85 to 100	100	--	--	--	90 to 100
10 mm	0 to 5	0 to 5	0 to 20	0 to 30	0 to 45	85 to 100	10 to 35	25 to 55	30 to 70	40 to 85
4.75 mm	--	--	0 to 5	0 to 5	0 to 10	0 to 20	0 to 5	0 to 10	0 to 10	0 to 10
2.36 mm	--	--	--	--	--	0 to 5	--	--	--	--

TABLE – 10

GRADING OF FINE AGGREGATES

[IS 383 – Table 4]

IS Sieve Designation	Percentage Passing for			
	Grading Zone I	Grading Zone II	Grading Zone III	Grading Zone IV
10 mm	100	100	100	100
4.75 mm	90 to 100	90 to 100	90 to 100	95 to 100
2.36 mm	60 to 95	75 to 100	85 to 100	95 to 100
1.18 mm	30 to 70	55 to 90	75 to 100	90 to 100
600 micron	15 to 34	35 to 59	60 to 79	80 to 100
300 micron	5 to 20	8 to 30	12 to 40	15 to 50
150 micron	0 to 10	0 to 10	0 to 10	0 to 15

NOTE – 1 For crushed stone sands, the permissible Limit on 150 micron IS Sieve is increased to 20 percent. This does not affect the 5 percent allowance permitted in 4.5 applying to other sieve sizes.

NOTE – 2 Fine Aggregate complying with the requirements of any grading zone in this table is suitable for concrete but the quality of concrete produced will depend upon a number of factors including proportions.

NOTE – 3 Where concrete of high strength and good durability is required, fine aggregate conforming to any one of the four grading zones may be used, but the concrete mix should be properly

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designed. As the fine aggregate grading becomes progressively finer, that is, from Grading Zones I to IV, the ratio of fine aggregate to coarse aggregate should be progressively reduced. The most suitable fine to coarse ratio to be used for any particular mix will, however, depend upon the actual grading, particle shape and surface texture of both fine and coarse aggregates.

NOTE – 4 It is recommended that fine aggregate conforming to Grading Zone IV should not be used in reinforced concrete unless tests have been made to ascertain the suitability of proposed mix proportions.

TABLE – 11

ALL-IN-AGGREGATE GRADING

[IS 383 – Table 5]

IS Sieve Designation	**Percentage for ALL IN Aggregate of**	
	40 mm Nominal Size	**20 mm Nominal Size**
(1)	**(2)**	**(3)**
80 mm	100	--
40 mm	95 to 100	100
20 mm	45 to 75	95 to 100
4.75 mm	25 to 45	30 to 50
600 micron	8 to 30	10 to 35
150 micron	0 to 6	0 to 6

TABLE – 12

MOISTURE CONTENT & BULKING PERCENTAGE RELATION FOR SAND

Moisture - Content % by weight	Bulking & (Volume)
2	15
3	20
4	25
5	30

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2.9 MISCELLANEOUS MATERIALS

A. Cover blocks shall be of non-corrosive material such as plastic but not wooden or broken bricks or stone. High-density PVC chairs specifically made for this purpose shall be used in the Work. Concrete cover spaces may be permitted by the ER. Such concrete spacers shall be cast from concrete and not cement mortar. Strength of these blocks shall be at least equal to the strength of concrete in use. These should be fully cured prior to use in the Work.

B. Binding wire shall be 16 or 18 gauge galvanized wire conforming to IS 280 . Binding shall be done with double wire. It shall be free from rust, oil, paint, grease, loose mill scale or any other deleterious material undesirable for the reinforcement and concrete or which may prevent adhesion of concrete with reinforcement.

C. Deformed bars for concrete reinforcement and rolled mild steel and medium tensile steel conforming to IS 1139 shall be allowed in the Work provided they are approved by the ER.

D. Grout: Free of gas-producing or gas-releasing agents, oxidizing catalysts, inorganic accelerators and chlorides. If requested by the ER, grout a sample (1.2m) x(1.2m) x (50mm) plate in the field and demonstrate a minimum of 95% bearing after curing 14 days. Cure grout in accordance with manufacturer's instructions.

E. Vapor Barrier: Maximum (2.3 ng/Pa•s•m^2) water vapor permeance. Include manufacturer's recommended adhesive or pressure-sensitive joint tape. ASTM E1745, Class A or equal.

F. Dustproofer and Chemical Hardener: Colorless, aqueous solution containing a blend of magnesium fluosilcate and zinc fluosilcate combined with a wetting agent, containing not less than (0.9kg/3.8L)of fluosilcate

G. Waterstops: Self-Expanding Strip Waterstops: Flexible strip of bentonite (granular bentonite clay (sodium bentonite), minimum 85% montmorillonite (hydrated aluminum silicate), with a minimum 90% passing a (0.03mm) sieve) waterproofing compound in cartridge or coil form, designed specifically for vertical and horizontal joints in concrete construction.

2.10 CURING MATERIALS

A. ASTM C1315. Water-based acrylic coating suitable for concrete curing and surface sealing. Clear transparent, low odor, high solids content. Section 12.5 of SP 24 shall also be followed.

2.11 REQUIREMENTS FOR MASS CONCRETE

A. Mass concrete clauses shall apply to concrete used in Mat Foundations and should be considered for footings greater than 1.0 m thick. Same materials and proportioning as normal weight structural concrete, except use moderate heat of hydration, moderate sulphate resistance cement, water reducing retarding admixture, and crushed ice if necessary to maintain required temperature.

1. Retardation: Minimum 3 hours.
2. Temperature: Maximum 21° C at point of deposit.

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2.12 REINFORCEMENT STEEL FABRICATION

A. General

1. Reinforcement shall be fabricated per the Drawings. Bending shall be done mechanically with the use of machines or, if approved, by hand but to the correct dimensions and radii, with proper tools and platforms and shall conform to IS 2502. Material shall be bent using cold bending only. Material shall be inspected for visible defects such as cracks, excessive rust, loose mill scale etc. Cracked ends of bars shall not be used in Work. Also, the bars should be free from any deleterious material and hence the best practice shall be to hose down reinforcement just prior to concreting.

2. It is important that bending, straightening, cutting etc. of bars be carried out in a manner that does not result in damage to the material, and that the safety of the persons working on rebar fabrication is ensured.

3. Anchoring of bars and stirrups shall be provided exactly as detailed in the structural Drawings or as directed by the ER..

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PART 3 - EXECUTION

3.1 GENERAL

A. Manufacturer's Instructions: Prepare substrates, apply primers and install the Work of this Section, including equipment, components, and accessories in accordance with the manufacturer's instructions, except where more stringent requirements are shown or specified, and where project conditions require extra precautions or provisions to ensure satisfactory performance of the Work

3.2 EXAMINATION

A. Verification of Conditions: Examine the areas to receive the Work and the conditions under which the Work would be performed. Contractor shall remedy conditions detrimental to the proper and timely completion of the Work. Do not proceed until unsatisfactory conditions have been corrected and approved by the ER.

3.3 PREPARATION

A. Substrate Acceptability: Commencement of installation shall constitute acceptance of substrate conditions by the Contractor.

3.4 FORMWORK

A. General: Design, erect, support, brace, and maintain formwork to support vertical and lateral, static and dynamic loads that might be applied until concrete structure reaches specified characteristic strength and can support loads. Forms shall be well constructed, carefully aligned, substantial, firm, securely braced and fastened together in their final position and set to give the finished structure the specified camber, plus allowance for shrinkage and settlement.

1. Forms shall conform to the lines, dimensions and shapes of concrete shown providing for openings, offsets, embedments, reglets, chamfers, blocking, screeds, anchorages, inserts, recesses, keys, slots, beam pockets, projections and other features as required.

2. Keep forms clean and free of foreign material before placing concrete. Provide selected materials to obtain required finishes in exposed Work. Solidly butt joints and provide backup at joints to prevent leakage of cement paste.

3. Provide openings in concrete formwork to accommodate Work of other trades. Determine size and location of openings, recesses, and chases from other trades providing items. Accurately place and support items built into forms.

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4. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.

5. Do not use earth cuts as forms for vertical surfaces, unless approved by the ER.

6. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment

B. Design of Formwork

1. The design and construction of the formwork shall be the responsibility of the Contractor. The drawings and calculations for the design of the formwork shall be submitted to the ER for information. Design of formwork shall take account of safety and surface finish. The formwork shall be sufficiently rigid and tight to prevent loss of grout or mortar from the fresh concrete.

2. Formwork shall be designed to withstand the worst combination of self-weight, reinforcement weight, wet concrete weight, concrete pressure, construction, seismic and wind loads together with dynamic effects caused by placing, vibrating and compacting the concrete. Forms shall be designed and constructed to maintain rigidity throughout the placing, ramming, vibration and setting of the concrete to the required shape, position and level and specified class of finish within the allowable tolerances.

3. All joints shall be sufficiently tight to prevent leakage of grout. If movement or deflection of the formwork or loss of grout occurs, the damaged concrete supported by such formwork shall be removed and the concrete re-cast so that the required finish is obtained. Formwork and its supports shall be designed to withstand the worst combinations of self-weight, reinforcement and wet concrete weights, concrete pressure, construction and wind and seismic loads. Due regard shall be taken of the type of mix when considering the design pressure on the formwork. The formwork shall be pre-cambered by an amount as shown on the Drawings.

4. If timber forms are used, they shall be of sound, well-seasoned timber free from loose knots. The forms shall be faced to give the specified class of finish for the structures. The formed surfaces of exposed concrete shall be smooth, true and free from all irregularities.

5. For below ground concrete except where against existing structures, rough formwork, steel pans etc., shall be used provided that all joints prevent the loss of grout.

6. Provide forms such that no discernible imperfections are in evidence in finished concrete surfaces due to deformation, bulging, jointing, or leakage of forms. Design forms to withstand the action of mechanical vibrators.

7. Design formwork to be readily removable without impact, shock or damage to cast-in-place concrete surfaces and adjacent structures or surfaces. Provide crush plates or wrecking plates where stripping may damage surfaces. Provide top forms for inclined surfaces where slope is too steep to place concrete with bottom forms only. Provide

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kerf wood inserts for forming keyways, reglets, recesses and the like to prevent swelling and to provide for easy damage free removal.

C. Tolerances: Measure forms before and during concreting and make appropriate adjustments.

1. Variation from Plumb
    a. In any (3m): . (6mm).
    b. In entire length/height: (19mm).
    c. In (6 m): (1:500)
    d. In (12 m): or more: maximum (1:500).
2. Variation from Level or Grades Indicated:
    a. Top Surface: +/- (19mm).
    b. Formed Surface: +/- (19mm).
    c. Top of Wall: +/- 19mm.
3. Variation from Building Lines
    a. In any . (3m): . (6mm).
    b. In entire length/height: (19mm).
4. Cross-Sectional Dimensions
    a. (6mm), + (13mm).
5. Slab openings: Variations in the sizes and locations of sleeves and slab openings shall not exceed (6mm)

D. Preparation of Form Surfaces

1. Use non-staining mineral oil or form lacquer.

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E. Form Construction

1. Construct forms to the exact sizes, shapes, lines, and dimensions shown, and as required to obtain accurate alignment, location, grades, level, and plumb Work in finished structures.

2. Provide for openings, offsets, keyways, recesses, moldings, reglets, chamfers, blocking, screeds, bulkheads, anchorages, inserts, and other features required. Use selected material to obtain required finishes.

3. Forms for openings, and construction which accommodates installation by other trades whose materials and products shall be fabricated before the opportunity exists to verify the measurements of adjacent construction which affects installations, shall be accurately sized and located as dimensioned on the Drawings. In the event that deviation from the drawing dimensions results in problems in the field, the Contractor shall be responsible for resolution of the conditions as approved by the ER's, without additional expense to the Employer.

4. Provide temporary openings where interior area of formwork is inaccessible for cleanout, for inspection before concrete placement, and for placement of concrete. Brace temporary closures and set tightly to temporary openings on forms in as inconspicuous locations as possible, consistent with design requirements. Form intersecting planes to provide true, clean cut corners.

5. Erect falsework and supports, brace and maintain it to safely support vertical, lateral, and asymmetrical loads applied until loads can be supported by in-place permanent construction. Construct falsework so that adjustments can be made for take-up and settlement.

6. Provide wedges, jacks, or camber strips to facilitate vertical adjustments. Carefully inspect falsework and formwork during and after concrete placement operations to determine abnormal deflection or signs of failure; make necessary adjustments to produce Work of required dimensions.

7. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

F. Form Stripping And Reshoring

1. General: Do not remove formwork or shoring until the concrete has obtained at least 75% of design strength and not until members are capable of supporting their own weight and construction loads without deflection exceeding 1/360 of span nor before the following minimum age requirements, unless otherwise noted.

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a)	Walls, columns and vertical faces of all structural members	24 to 48 hours as may be decided by the ER.
b)	Slabs (props left under)	3 Days
c)	Beam soffit (props left under)	7 Days
d)	Removal of props under slabs : 1) Spanning up to 4.5 m 2) Spanning over 4.5 m	 7 Days 14 Days
e)	Removal of props under beams and arches : 1) Spanning up to 6 m 2) Spanning over 6 m	 14 Days 21 Days

2. For cements other than Ordinary Portland Cement, the stripping time recommended shall be suitably modified. The number, size and position of props left under shall be such as to be able to safely carry the dead load of the slab, beam or arch, together with any live load likely to occur during curing or further construction.

3. Sleeves for through bolts shall not be provided in formwork for liquid retaining structures as they are potential sources for leakage. Special devices shall be used for tying and adjusting forms and shuttering. Surface depressions at devices shall be filled with patching mortar.

G. Surface Finishes from Formwork: Generally formwork shall be specified as either wrought or unwrought depending on the required surface finish.

1. Unwrought Formwork

   a. Unwrought Formwork shall consist of sawn boards, brick or concrete block work, sheet metal or other suitable material to give adequate support to the concrete.

   b. Appearance is not of primary importance for this class of formwork.

   c. Surfaces to which plaster, granolithic or other finish is to be applied shall be roughened while the concrete is still green.

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2. Wrought Formwork

a. Wrought formwork shall be provided for concrete surfaces that are required to be finished smooth. Wrought formwork shall be lined with metal or plywood having smooth surfaces and edges.

b. Formwork shall be furnished in largest practicable sizes to minimize the number of joints. Care shall be taken to ensure that there are no irregularities or rough areas between successive sections of shuttering so that finished surfaces shall be free of board or shutter marks. Upon removal of formwork, surfaces of finished concrete shall be rubbed down with carborundum stone where necessary to obtain a uniform and smooth appearance.

c. The finish shall be such as to require no filling of surface pitting, butt fins, surface discoloration and other minor defects shall be remedied by ER approved methods.

d. Rendering of defective concrete as a means of making good will not be permitted except that, in case of minor porosity on the surface, approval may be given for the surface to be treated by rubbing down with a cement mortar of the same fine aggregate/cement ratio as the concrete. The treatment shall occur immediately after removing the formwork. Both cement and aggregate shall be from the same source as the concrete materials.

e. Concrete containing honeycombing, major air holes or similar defects shall be cut out and replaced as directed. No repair shall be executed without ER approval.

H. Ties

1. Where it is required to use internal ties and spacers, their type, spacing and use shall be approved by the ER. No part of any such tie or spacer remaining permanently embedded in the concrete shall be nearer than the specified cover to the finished surface of the concrete. Wire ties projecting through the concrete face shall not be permitted.

I. Permanent Formwork.

1. Permanent formwork for elevated concrete floors in steel structures shall be profiled steel sheet. The sheet shall be of sufficient thickness to sustain all construction loads plus the weight of fresh concrete between supporting beams without excessive deflection. The underside of the sheet shall be coated with an approved corrosive resistant paint. The material shall conform to IS513 (D) / IS 10748 Gr.2.

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2. Permanent form deflection shall be limited to span / 360. The sheet shall be fixed to the supporting beams at every alternate valley using 10 mm dia fusion welding with 22 mm dia x 2.5 mm thick reinforcing washers. Minimum 100 mm seating shall be provided. Deck sheet ends resting on walls shall have a seating of 150 mm minimum. Side laps shall be secured with tack welds to ensure that the slurry does not leak out. End laps shall always be on the supporting walls or on supporting beams. The overhang of the deck sheet shall be limited to 300 mm maximum. Temporary supports, wherever required, shall be firm, unyielding and at the same level as the permanent supports and shall be in place till the concrete attains its full strength. While removing the temporary supports, care shall be taken that the slab is not disturbed.

J. Formwork for Vibrated Concrete

1. If external vibrators are to be used for compacting the concrete, appropriate type of vibrator, design of the formwork and the method of fixing the vibrators shall be used.

K. Cleaning and Treatment of Forms

1. Before concreting is commencing the forms and previously cast concrete shall be thoroughly cleaned and free from all sawdust, tie wire, shavings, dust, dirt and other debris. Temporary openings shall be provided where necessary to drain away water and remove rubbish.

2. All rubbish shall be removed from the interior of forms before the concrete is placed. The faces of forms in contact with the concrete shall be cleaned and treated with a suitable release agent. The release agent shall be applied so as to provide a thin uniform coating to the forms without contaminating the reinforcement.

3. Release agents shall be applied and be compatible with the class of finish. Care shall be taken not to contaminate the reinforcement.

L. Reshoring: Provide immediately after stripping slabs, beams, and girders that support subsequent formwork. Retain reshores for as many levels as required to combine the live load capacities of cured floors to support the loads of the subsequent fresh construction and construction loads. Reshore a minimum of 2 consecutive stories below the story of primary formwork shoring.

3.5 REINFORCEMENT

A. General: Clean reinforcement of loose rust and mill scale, earth, and other materials that reduce or destroy bond with concrete. Where vapor barrier is required, avoid cutting or puncturing vapor barrier during reinforcement placement and concreting operations.

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B. Position, support, and secure reinforcement against displacement by formwork, construction, or concrete placement operations. Locate and support reinforcing with metal chairs, runners, bolsters, spacers, and hangers, as required. Place reinforcement to obtain the required minimum coverages for concrete protection. Arrange, space, and securely tie bar supports together with (1.6mm) wire to hold reinforcement accurately in position during concrete placement operations. Set wire ties so that twisted ends are directed away from exposed concrete surfaces.

C. Place reinforcing bars having assigned positions so that distinguishing marks agree with those given on the shop drawings relating to or calling for the bars. Provide supports of sufficient number and strength to carry reinforcement. Do not place reinforcing bars more than (50mm) beyond the last leg of any continuous bar support. Do not use supports as bases for runways for concrete conveying equipment or similar construction loads.

D. Lapping of Bars

1. Laps shall be strictly as per the Drawings or as directed by the ER. For general guidance, the following principles shall be followed as given in IS 456.

    a. Splices shall be provided as far as possible away from sections of maximum stress and be staggered.

    b. Not more than half of the total bars shall be spliced at a section.

    c. Where more than one half of the bars are spliced at a section or where splices are made at points of maximum stress, special precautions shall be taken, such as increasing the length of lap and / or using spirals or closely spaced stirrups around the length of the splice.

    d. Lap splices shall not be used for bars larger than 36 mm diameter. For larger diameters, bars may be welded. In cases where welding is not practical, lapping of bars larger than 36 mm diameter may be permitted and additional spirals should be provided around the lapped bars.

    e. When splicing of welded wire fabric is to be carried out, lap splices or wires shall be made so that the overlap measured between the extreme cross wires shall be not less than the spacing of cross wires plus 100 mm.

E. Spacing of Bars

1. Bars shall be placed in position as shown in the Drawings. The following guidelines as given in IS 456 shall be followed in case of difficulties or shall be carried out as directed by the ER.

    a. Horizontal distance between two parallel main reinforcing bars shall usually not be less than the greatest of the following.

        1) The diameter of the bars, if the diameters are equal.

        2) The diameter or larger bar, if the diameters are unequal, and

        3) 5 mm more than the nominal maximum size of coarse aggregate (by

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using reduced size of aggregate in congested reinforced area, conditions given hereof should be overcome)

2. Greater horizontal distance should be provided. But when needle vibrators are used, distance between bars of a group may be reduced to two-third of the nominal maximum size of the coarse aggregate, provided sufficient space is left between groups of bars to enable the vibrator to be immersed.

3. Where there are two or more rows of bars, the bars shall be vertically aligned and the minimum vertical distance between the bars shall be 15 mm, two third the nominal maximum size of the aggregate or the maximum size of bar, whichever is greatest.

F. Cover to Reinforcement

1. Reinforcement shall have concrete cover and the thickness of such cover shall be as specified in the Drawings or as directed by the ER.

G. Fixing in position

1. Correctly cut and bent bars shall be accurately placed in position as detailed in the Drawings. Unless otherwise specified by the ER, reinforcement shall be positioned within the tolerance as under:

   a. for effective depth 200 mm or less ± 10 mm

   b. for effective depth more than 200 mm ± 15 mm

2. But in no case shall the cover be reduced by more than 5 mm of that specified. There shall be no compromise on cover for foundation work.

3. Reinforcing bars shall be held in position during the placing of concrete by use of PVC or concrete cover blocks (made of equal / of higher grade strength of well-cured concrete in use) steel chair spacers, steel hangers, supporting wires, etc. and secured by tying with an annealed binding wire of 16 to 18 gauge as approved by the ER.

4. Layers of bars shall be separated by precast concrete spacer blocks or spacer bars. Reinforcement shall be in correct position prior to start of concreting. No reinforcing bar shall be placed on freshly laid concrete for adjusting bar spacing. Care shall be taken to maintain reinforcement in position and keep it clean, throughout the period till it is embedded in the concrete..

5. Binding wire used shall conform to IS 280.

H. Welded joints or mechanical connections

1. Welded joints or mechanical connections in reinforcement may be used but in all cases or important connections, tests shall be made to prove that the joints are of the full strength of the connected bars. Welding of reinforcement shall be done in accordance with IS recommendation.

2. Where reinforcement bars are bent aside at construction joints and afterwards bent

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back into their original position, care shall be taken to ensure that at no time is the radius of the bend allowed to be less than 4 bar diameters in case of plain mild steel or 6 bar diameters for deformed bars. Care shall be taken when bending back bars to ensure that the concrete around them is not damaged / disturbed.

3. Welding rods used shall conform to IS 814: covered electrodes for metal arc welding of structural steel. Work shall be carried out by a certified welder. Samples from Work site shall be taken at regular intervals and tested. Frequency and number of samples shall be as directed by the ER.

I. Unacceptable materials: Reinforcement with any of the following defects shall not be permitted in the Work:

1. Bar lengths, depths, and bends exceeding specified fabrication tolerances.

2. Bends or kinks not indicated on Drawings or final Shop Drawings.

3. Bars with reduced cross-section due to excessive rusting or other causes

J. Restore damaged bars to full capacity in a manner acceptable to the ER.

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3.6 CONCRETE

A. Concrete is prepared by mixing graded aggregates along with cement, water and approved admixtures, in a specified proportion. Mixing shall be done using a mechanical mixer. Manual hand mixing shall be permitted in specific cases with the written permission of the ER on account of small quantity or location or any other reason acceptable to the ER.

B. Cement Concrete.

1. This shall be classified as plain cement concrete or reinforced cement concrete. Plain cement concrete shall be used in leveling courses under foundations, floors, copings etc. Reinforced cement concrete shall be used typically at all locations.

a. Concrete shall be classified by its compressive strength at the 28th day. The concrete grades shall be as designated in table 2 of IS 456 and are given as ready reference in Table — 4.

b. It shall be the responsibility of the Contractor to carry out design mixes and approval of the same shall be obtained from the ER at least 35 days in advance from the actual pouring of concrete at site in the permanent Work. The basic aim of mix design shall be to find the most economic proportion of cement, aggregates and water which will give the desired strength of concrete, proper workability and durability. Also it is important that the mix should be easily worked with the help of equipment available at site. The operations involved at site are, measurement of materials, their mixing, placing, compacting, finishing required and curing. The design shall be carried out strictly to IS specifications and IS code practice 456, SP 23 and SP 24.

c. The Contractor shall ensure that the minimum cement content per cubic

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meter of reinforced concrete should not be less than that stipulated in table 23, 24, 25 and 26 of SP 23 and Table 5 and Table 6 herein.

d. For expected strength of cubes tested on the 28th day the design mix at preliminary test and work site shall be as per IS 456, SP 23 and SP 24. The water cement ratio shall be between 0.5 to 0.52. Additional water may be permitted only at the discretion of the ER. The slump shall be 25 mm to 35 mm depending upon the location and type of Work. Higher slump with use of plasticizers shall be permitted.

C. Design Mix And Trial Mixes

1. The Contractor shall submit the mix designs that he proposes to use at site to the ER for review at least 5 weeks in advance of planned use, The mix design shall also give basic details, when tested according to Is 1199 and IS 2386 – Part III, such as.

   a. Slump

   b. Vee - Bee Test

   c. Bleeding

   d. Cement required for one cubic meter of concrete

   e. Compacting factor

2. On receipt of this, the ER may immediately order a work site test to be performed before the final approval. This shall be done with mixer and materials actually being used at site.

3. 5 cubes shall be taken from each of the 3 batches to test the mix. Cubes shall be cast, stored, cured, transported and tested per IS 516. The test may be carried out at site or laboratory as approved by the ER.

4. Trial mixes shall be approved provided that average strength of 3 consecutive cubes is not less than that specified and that one out of three may give a value less than specified but limited to a maximum of 90% of the specified strength.

5. In case the trial mix falls below the above criteria, the ER shall order fresh trial mixes to be made as before, until the desired strength is arrived at.

6. This design mix and trial mix hold good so long as the materials continue to be of the same quality and from the same sources. For any change, the ER may order fresh design mix and trial mixes to be carried out before the same is used at site.

7. It is the responsibility of the Contractor to prepare and get the cubes tested and to provide all the material, labour, moulds, equipment, casting and curing required.

8. The Contractor shall provide and maintain all the equipment and staff at the site throughout to carry out the following tests in an approved testing laboratory, hereinafter referred to as the Contractor's laboratory.

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a. Grading of coarse and fine aggregates.

b. Silt content of sand.

c. Moisture content of coarse and fine aggregates.

d. Slump.

e. Concrete cube strength.

9. The Contractor shall maintain full records of all above tests in a register.

10. The format of records shall be prepared in consultation with the ER and either he or his representative shall have full access to the Contractor's laboratory..

D. Mixing Of Concrete

1. Weight batching shall be preferred at site but subject to the ER's approval it may be permit designed mixes to be converted to volumetric equivalents if requested in writing by the Contractor on specific grounds. The Contractor shall provide required boxes to measure the ingredients of concrete.

2. The Contractor shall provide concrete batch mixers, vibrators, weight batches conforming to relevant IS specification. The capacity and number of mixers and vibrators required at site from time to time shall be to the approval of the ER. No equipment from site shall be removed without the prior written approval of the ER. The Contractor shall also maintain a platform weighing scale of capacity 300 kg with fraction upto 100 gms at the site.

3. As directed by the ER, a weekly or periodic calibration of all machines shall be done and records of these calibrations shall be maintained in a register. Regular maintenance of machinery shall also be carried out on a weekly basis or as directed by the manufacturer of machines.

4. The mixer shall be run for a minimum period of 2 minutes after all materials are loaded in full quantity. The concrete produced shall be uniform in colour and consistency.

   a. Excessive mixing requiring the addition of water to preserve the required consistency shall not be permitted. Mix concrete to a consistency which can be readily placed without segregation.

   b. Equip mixers with a device for measuring and dispensing admixtures. When required, the slump of concrete may be increased prior to placement by the use of a high range water reducing admixture.

5. Hand-Mixed Concrete: When hand-mixed concrete is allowed and approved for certain parts of the Work, mix on watertight platforms. Proportion cement, sand and aggregate loose by volume, carefully measured. Thoroughly mix sand and cement together dry until the mixture is of uniform color. Add the aggregate and turn the

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mass over until the mixture is uniform and homogeneous. Add water by sprinkling and turn the mass over until it is uniformly mixed and of the required consistency.

6. Retempering: No water shall be added to the mix after the initial batching operation. Concrete which is placed after the unauthorized addition of water is subject to removal.

7. When required, the slump of concrete as delivered to the site may be increased prior to placement by the use of a high range water reducing admixture. Dosage of superplasticizer shall be within the guidelines established by the manufacturer and agreed to by the ER. Alternative Mix Design Reports with the recommended site dosage(s) of superplasticizer shall be submitted prior to use.

8. The placing temperature of concrete shall not be more than 34° C. If it is more, the ER may order addition of ice or chilled water to the concrete at no additional cost to the Employer. Also the Contractor shall take the following precautions:

   a. Mixers and weight batches shall be painted with white colour

   b. Aggregate storing bins shall not be exposed to the Sun.

   c. Water shall be sprinkled on aggregates well before concreting to keep the temperature low.

E. Shrinkage cracks

1. Concreting shall be avoided in very warm weather, if necessary, it shall be covered with dampened Hessian within 2 hours of placing of concrete.

2. To achieve good results the concrete shall be immediately covered with a plastic sheet and not exposed to any direct wind contact.

F. Laying of Concrete

1. Concreting shall commence only after formwork is approved, reinforcement is recorded and permission to proceed with concreting has been approved in writing from the ER.

2. Formwork should be clean, free from sawdust, pieces of wood, bottles, or any other foreign material. Coat contact surfaces of forms with an approved, non-residue forming form-coating compound before reinforcement is placed. Do not allow excess form coating material to accumulate in forms or to come into contact with in-place concrete surfaces against which fresh concrete will be placed. Apply in compliance with manufacturer's written instructions and recommendations. It should have been treated by form releasing agent prior to the laying of reinforcement and concrete.

3. Concrete shall be as gently deposited as is practically possible. In its final position to avoid re-handling and shall be so deposited that segregation of aggregates does not occur. In the case of deep trenches and footings, this may be done with the help of a chute.
   Columns and walls shall be so adjusted in formwork so that maximum fall is 1.5

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meters unless consented to by the ER. Concrete from wheelbarrows shall not be dumped away from the face concrete already in place. It shall be dumped into the face of concrete already in place.

4. Concrete placed onto a sloping surface shall be discharged by providing a chute with a baffle and a drop at its end so that the concrete remains on the slope.

5. Columns and walls shall be concreted in one operation to their full height to avoid any horizontal construction joints as far as possible.

6. All slabs, beams, wooden planks and cat-walks shall be provided clear of reinforcement. Concrete shall be placed in position within 30 minutes from the time it is produced. Concrete shall be laid during normal working hours. Concreting at night or on holidays shall be permitted only on the written approval of the ER.

7. Free Fall: Guide the flow of concrete in walls and columns for vertical drop between the reinforcing. Limit the free fall of concrete to maximum 2.5.m.

G. Compaction of Concrete

1. Concrete shall be thoroughly compacted, as depositing shall proceed by means of suitable vibrators. The vibrators shall maintain the entire concrete under treatment in an adequate state of agitation that shall continue during the whole period of placing of concrete. Care shall be taken not to over- vibrate the concrete. While withdrawing needles no holes shall be visible in concrete. Compacting shall be completed before the initial setting time. Concrete already set shall not be disturbed by successive vibrations.

2. Insert and withdraw vibrators vertically drawing out entrapped air and excess water. Do not use vibrators to transport concrete within forms. Do not insert vibrators into lower layers of concrete that have begun to set. At each insertion limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing segregation of mix. Pay particular attention to providing adequate vibration in the vicinity of sleeves and embeds in heavily reinforced elements and other elements as required. Provide form vibrators to ensure maximum density of concrete.

3. It shall be ensured that the needle vibrators are not applied to reinforcement, which may destroy the bond between concrete and reinforcement. When electric vibrators are in use, a standby petrol vibrator shall always be available at the concreting point.

H. Construction joints

1. In large pours, it is practically not possible to carry on concreting continuously. Hence construction joints shall be provided. Location of construction joint shall be submitted by the Contractor for approval of the ER. Such joints shall be kept to a minimum. The joints shall be at places where shear forces are nil or minimum and these shall be straight and at right angles to the direction of the main reinforcement.

2. Form construction joints with a shear key. In walls, do not space construction joints more than 18m apart in horizontal direction, maximum 6m apart in vertical direction

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unless otherwise shown. In framed slabs and beams, and on metal deck slabs, do not space construction joints more than 27m apart. In slabs on grade, do not space construction joints more than maximum 18m in any direction. Place construction joints perpendicular to main reinforcement. Continue reinforcement across construction joints except as otherwise shown.

3. Stop ends provided shall have slots for reinforcement bars to pass feely without bending or any other obstruction. Also, a trapezoidal fillet nailed on a stop board shall be provided to form a regular keyed joint. Joints shall be straight and truly vertical or horizontal. Before commencement of concrete placement, adjacent concrete stoppers and surfaces shall be chipped and roughened to expose aggregate, then wire brushed and cleaned. The concrete surface shall be sprayed with water for 24 hours before casting and kept wet until casting. True horizontal joints shall also be provided with a keyed joint by inserting planed greased timber. This shall be treated as above prior to the start of fresh concreting.

4. For vertical joints neat cement slurry shall be applied on the surface just before concreting. For horizontal joints, the surface shall be covered with a layer of mortar about 10 to 15 mm thick composed of cement and sand in the same ratio as the cement and sand in the concrete mix. This layer of cement slurry shall be freshly mixed and applied just before concreting.

I. Expansion Joints

1. Expansion joints shall be formed and located as detailed in the Drawings.

2. Install joint filler to allow the required dimension for sealant, as indicated. Dimensions shown on the Drawings are based on an assumed design temperature of 21 deg. C Concreting procedures shall take into account the ambient temperature range at the time of the respective operations. Do not stretch, twist, puncture, tear, or leave gaps between ends of joint fillers.

J. Waterstops:

1. Self-Expanding Strip Waterstops: Install in construction joints and at other locations as indicated, according to manufacturer's written instructions and recommendations, bonding or mechanically fastening and firmly pressing into place. Install in longest lengths practicable.

K. Embedded Items and Inserts:

1. Accurately set anchorage devices by line and transit, and coordinate the locating of anchorage devices to be set for the accommodation of the Work of other trades.

2. Locate anchor bolts as shown on the Drawings and on shop drawings. Obtain necessary templates from the mechanical trades as required for the proper setting of anchor bolts and other items for mechanical equipment, as required.

3. Assist other trades in the installation of piping, pipe sleeves, conduit and similar items where items are to be installed in concrete. Provide frames to securely hold anchor bolts and anchorage devices in place during construction, and take care that no

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displacement occurs during the pouring of concrete. Under this Section furnish and set items not furnished by other trades using approved standard type items suitable for their intended purpose.

4. Inserts and embedments shall located as shown in the Drawings and shall be protected from weather and damage in course of the construction.

L. Openings

1. All required openings and block-outs (pockets) shall be provided as detailed in the Drawings. All openings and pockets shall be de-shuttered with care and all corners of openings shall be preserved. All openings / block-outs (pockets) shall be in a correct line and level. After concreting, the openings shall be secured against any accident by proper coverings, guardrails and warning notices, as appropriate.

2. The Contractor shall clean and grout block-outs / pockets at a later date with a non-shrinking grout mix or concrete of matching strength. They shall be well-cured and protected to correct line and level till handling over

M. Filling-In: Fill in block-outs, holes and openings left in concrete structures for passage of Work by other trades, unless otherwise shown or directed, after Work of other trades is in place. Mix, place, and cure concrete as herein specified, to blend with in-place construction. Use non-shrink materials or admixtures as appropriate. Provide other miscellaneous concrete filling shown or required to complete the Work.

N. Curbs: Provide monolithic finish to interior curbs by stripping forms while concrete is still green and steel-troweling surfaces to a hard, dense finish with corners, intersections, and terminations slightly rounded.

O. Equipment Bases and Foundations: Provide machine and equipment bases and foundations, as shown. Set anchor bolts for machines and equipment to template at correct elevations, complying with certified diagrams or templates of manufacturer furnishing machines and equipment.

P. Curing

1. After concrete has begun to harden i.e. about 1 to 2 hours after placement, it shall be protected from quick drying with moist or damped Hessian cloth or any other material approved by the ER. Starting 24 hours after placement of the concrete, the surface shall be cured by flooding with water or covering with damp Hessian cloth for a period of 7 days to keep it moist.

2. For the next 7 days the surface shall be kept wet all the time by sprinkling water continuously.

3. For Mass Concrete: Curing period shall be 14 days.

4. For membrane curing, details as listed in 12.5 of SP 24 shall be followed.

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3.7 FINISHES FOR FORMED SURFACES

A. Application of Formed Finishes: The areas of each finish are designated on the Drawings. For guidance the following points shall be noted but the ER shall be consulted prior to start of concreting and his decision in this regard shall be final.

B. Moist cure in the forms. Do not sand float, bag, sack, grout clean, or otherwise apply cementitious masking to formed surfaces.

1. Related Unformed Surfaces: Screed, float and trowel unformed surfaces to produce texture consistent with adjacent formed surfaces or as indicated on the Drawings.

2. Patching: Correct minor surface defects and fill holes, unless otherwise noted. Use a mortar consisting of 1 part Portland cement to 1 part fine aggregate by volume and apply over a bonding agent, or use a proprietary patching compound.

C. Corner Protection: Protect horizontal and vertical corners of columns and walls for full length or full height of exposed corner with continuous wood corner guards. Place wood corner guards immediately after stripping of forms. Remove and replace guards as required after treatment and maintain until possibility of damage due to construction operations no longer exists.

D. Concealed Formed Surfaces: Remove fins, and patch tie holes.

E. Exposed Formed Surfaces, Unpainted: Remove fins, patch tie holes, and stone the joint marks and out-of-plane surfaces to within 3 mm of flush. Rub with carborundum stone using only enough water to develop cement past from concrete mortar and to produce uniformly dense and smooth concrete.

F. Exposed Formed Surfaces, Painted: Remove fins, patch tie holes, and stone the joint marks and out-of-plane surfaces to within 1.5 mm of flush. Rub with carborundum stone using only enough water to develop a cement paste from the concrete mortar and to produce uniformly dense and smooth concrete.

G. Exposed Formed Surfaces, Smooth As-Cast Finish: Maintain concrete free from laitance caused by spillage, leaking forms or other contaminants. Do not allow laitance to penetrate, stain or harden on finished surfaces. Do not attempt surface patching or cleaning, if required, unless acceptable to Architect and ER. Patch tie holes to match the surrounding surfaces and strike of flush.

3.8 FINISHES FOR FLATWORK

A. General Requirements for Flatwork: Strike-off top surfaces of flatwork true and level to the required tolerances. Use construction techniques such as adjustment of pour size, adjustable screeds, pre-shoring/shoring/re-shoring and other appropriate means to ensure compliance with these requirements. Deflections of unshored structural steel and metal deck due to wet weight of concrete and structural steel camber are shown on the Drawings. Provide concrete, in addition to scheduled nominal slab thickness, to allow for the steel or formwork deflections and camber and provide finished slabs level within the specified requirements. Monitor and survey pour areas prior to, during and after concreting operations.

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B. Flatwork Finish Levelness Tolerances: Construct flatwork such that, for every 3m distance from a permanent point of support (column, bearing wall, etc.) the finished slab surface shall be no more than 6mm below a theoretical horizontal plane projected from that support.

C. Fill or grind completed floors as necessary to achieve specified finish tolerances. Filling, where required, shall be with a self-leveling cementitious product capable of being tapered to a feathered edge.

D. Troweled Finish: Level surface and remove excess laitance by tamping, screeding, and preliminary wood floating. When the slab has hardened sufficiently so that water and fine material will not be worked to the top, compact the surface with disc type motor-driven floats. Cut down high spots and fill low spots. Trowel smooth with two steel troweling operations. Do the second troweling after the concrete has become so hard that no mortar will adhere to the edge of the trowel, and exert heavy pressure to thoroughly compact the surface. Leave floors with a smooth, hard finish free of blemishes and true to the maximum flatness tolerance specified. Grind surface defects that would telegraph through applied floor covering system smooth. Trowel finish surfaces scheduled to receive the following:

1. Resilient flooring.
2. Adhesive-set tile.
3. Carpeting.
4. Epoxy Terrazzo
5. Where no other finish is specified.

E. Float Finish: Level surface and remove excess laitance by tamping, screeding, and preliminary wood floating. When the slab has hardened sufficiently so that water and fine material will not be worked to the top, compact the surface with disc type motor-driven floats. Cut down high spots, fill low spots and leave floors with a smooth finish and true to the maximum flatness tolerance specified. Float finish surfaces scheduled to receive the following:

1. Membrane waterproofing.

F. Unfinished Slab Finish: Tamp the concrete using special tools to force the aggregate away from the surface, then screed with straightedges to produce a true and uniform surface as specified herein. Provide unfinished rough slab finish surfaces scheduled to receive the following:

1. Tile (except adhesive-set).
2. Floor topping (Screeds).
3. Insulation.
4. Concrete fill.

G. Broomed Finish: Tamp the concrete using special tools to force aggregate away from the

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surface, then screed with straightedges to bring surfaces to the required lines. While the concrete is still green, wood float to a true and uniform plane with no coarse aggregate visible, and apply a broom finish, in the direction indicated. Broom finish the following:

1. Driveways.

2. Ramps.

3. Parking area slabs.

H. Interior Exposed Concrete Surfaces of Ceilings, Columns and Walls: Unless otherwise shown, leave a smooth finish, even-textured and free of blemishes. Repair or replace defective areas. As soon as the face forms are removed, remove fins and other projections carefully, level offsets and grind where necessary as soon as the face forms are removed. Repairing, replacing and pointing and filling voids shall be done to the ER's satisfaction.

I. **I**nspection and corrective measures

1. On removal of formwork, the surface shall be examined by the ER. Till such time, the Contractor shall carry out no remedial measures. All patching, rectification or chipping shall be done only on the ER's instructions. In case of any violation of this rule, the concrete poured stands rejected. The decision of the ER in this regard shall be final and binding on all parties.

2. Sagged, bulging, patched, honeycombed Work shall stand to be rejected for surfaces that are exposed, or require fair face finish or decorative textured finish. The ER may permit any Work found structurally safe and areas of unexposed faces to be repaired.

3. Cracks observed shall be brought to the notice of the ER who shall examine them. They shall be kept under observation and a record shall be maintained for a period of 45 days. It shall be shown to the ER and the following procedure shall be followed.

    a. Cracks not developing further and in the opinion of the ER not detrimental to the strength of the construction shall be grouted with non-shrinking cement slurry or as directed by the ER.

    b. Cracks developing further and, in the opinion of the ER, detrimental to the strength of construction, shall be tested as per the relevant Indian Standard as directed by the ER.

    c. Based on result of the test, the ER shall order remedial measures or order the Contractor to dismantle construction, remove the debris, replace the construction and carry out all the consequential Work thereto.

4. Quantity of test cubes and testing

5. The minimum frequency of cube casting for testing shall be as follows. Each sample shall consist of 6 cubes.

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Concrete quantity	**Number of Samples**.	
Up to 5 cu.m. in a day	15 cu.m. to 15 cu.m.	2
15 cu.m. to 30 cu.m.	3	
30 cu.m. to 50 cu.m.	4	
More than 50 cu.m.	4 + one additional per each 50 cu.m. or part thereof.	

Three cubes shall be tested on the 7th day and other three cubes on the 28th day.

6. For Mass Concrete, One additional sample shall be taken and reserved for testing at 56 days.

7. Acceptance of Work

   a. This shall be as given in IS 456, SP 23 and SP 24. Part or element of Work shall be deemed to be accepted, provided the results of the 28th day cube testing conform to the criteria stated as under

      1) The average of the three consecutive cube's strength shall not be less than the specified strength.

      2) No individual cube strength shall be less than 90% of the specified strength.

      3) If an individual cube strength is more than 33% below the specified strength, such a cube shall be classified as a freak and the criteria in (a) and (b) shall be applied to the remaining two cubes and their acceptability determined.

      4) If the concrete tests fail to meet the acceptance criteria of the minimum strength required for respective grades of concrete, the ER may take one of the following decisions:

      5) Instruct the Contractor to carry out such additional tests (e.g. core tests, load tests etc.) and / or remedial measures to ensure the soundness of the structure at the Contractor's expense.

         a) Any decision to accept the Work shall be entirely at the discretion of the ER who may order a reduction in the rate of the appropriate item.

         b) The Work will be rejected and any consequential action as needed shall be taken at the Contractor's expense including cutting out and replacing a part or whole of the Work.

3.9 FIELD QUALITY CONTROL

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A. Contractor's Quality Control Responsibilities: Contractor is solely responsible for quality control of the Work and shall develop and maintain a Quality Control Program for the Work.

B. Certification: Certify, by written report submitted on a weekly basis, for each level, that the elevations, finish lines and building lines of the formwork systems are within specified tolerances, as substantiated by transit survey and that embedments and inserts have been installed within tolerance. Perform survey at locations indicated by the ER (minimum one per bay per floor) of formwork immediately before casting of concrete.

3.10 INDEPENDENT TESTING

A. Independent Testing Agency: A testing agency, engaged at the Contractor's expense, will perform the activities listed herein.

B. Monitor the Contractor's Quality Control Program. The monitoring activities do not relieve the Contractor of sole responsibility for maintaining the Quality Control Program.

C. Monitoring of Concrete Placement: Continuously observe concrete placement operations on a full-time basis, record such observations on a daily basis, and submit reports of the results.

D. Monitoring of Contractor's Testing: Monitor Contractor's non-destructive strength testing for form removal and termination of curing.

E. Compression Tests: Sample at point of deposit. Cure the cylinders in laboratory.

1. Additional Cylinders: Furnish a supply of 25 standard cylinder molds at the site at all times for making additional cylinders as may be directed by the ER.

2. Quantity of test cubes and testing

a. The minimum frequency of cube casting for testing shall be as follows. Each sample shall consist of 6 cubes.

Concrete quantity	**Number of Samples**.
Up to 5 cu.m. in a day	1
5 cu.m. to 15 cu.m.	2
15 cu.m. to 30 cu.m.	3
30 cu.m. to 50 cu.m.	4
More than 50 cu.m.	4 + one additional per each 50 cu.m. or part thereof.

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b. Three cubes shall be tested on the 7th day and other three cubes on the 28th day.

c. For Mass Concrete, One additional sample shall be taken and reserved for testing at 56 days.

d. Acceptance of Work

This shall be as given in IS 456, SP 23 and SP 24. Part or element of Work shall be deemed to be accepted, provided the results of the 28th day cube testing conform to the criteria stated as under

1) The average of the three consecutive cube's strength shall not be less than the specified strength.

2) No individual cube strength shall be less than 90% of the specified strength.

3) If an individual cube strength is more than 33% below the specified strength, such a cube shall be classified as a freak and the criteria in (a) and (b) shall be applied to the remaining two cubes and their acceptability determined.

4) If the concrete tests fail to meet the acceptance criteria of the minimum strength required for respective grades of concrete, the ER may take one of the following decisions:

5) Instruct the Contractor to carry out such additional tests (e.g. core tests, load tests etc.) and / or remedial measures to ensure the soundness of the structure at the Contractor's expense.

a) Any decision to accept the Work shall be entirely at the discretion of the ER who may order a reduction in the rate of the appropriate item.

b) The Work will be rejected and any consequential action as needed shall be taken at the Contractor's expense including cutting out and replacing a part or whole of the Work.

e. Compression Test Reports: Present at least the following data in tabular form and distribute immediately after recording the test results.

1) Identity of job, Contractor, supplier.

2) Identity of mix and required strength.

3) Pour location of sampled concrete.

4) Slump, water content, air content, truck number, time and date sampled, air temperature, concrete temperature, consistency.

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5) Curing history.

6) Date tested.

7) Compressive strength.

8) Type of fracture.

9) Compliance with specifications (yes or no)

F. Slump Tests: First truck each day, each sample for cylinders, and as often as necessary thereafter. Provide a slump cone and rod available for use at all times. Verify water content in accordance with AASHTO T318, "Measurement of Water Content of Fresh Concrete Using the Microwave Oven" as required by the ER, but not less than one test per day.

G. Reinforcement Material Verification: Verify materials conform with the Contract Documents.

H. Reinforcing Placement Inspection: Periodically inspect the placement of steel reinforcing, embedments, and insert placement. Provide daily reports of compliance.

I. Formwork Inspection: Periodically observe formwork shape, location and dimensions and shoring operations; record such observations on a daily basis.

J. Independent Testing Agency Reports: Submit for ER's information. Furnish reports for the specified tests. Test reports indicating non-compliance should be immediately emailed to the ER.

END OF SECTION

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